

GIZMO Ground-Reference Monitoring Operations Guide

Connection, commissioning, and monitoring during construction and operation
DUNE Far Detector

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Prepared from the available DUNE DocDB, ORC, and legacy operating record

Public review update of August 18, 2026

STOP / SAFETY: This is an integration and commissioning guide, not an approved electrical work procedure and not a substitute for the current Far Detector grounding drawings, ORC, LOTO, hazard analysis, or Procore SOP004. Fixed grounding conductors, the saturable-inductor safety path, the current transformer, and welding returns may be installed or changed only by authorized electrical personnel under the approved work plan. **Never lift the safety ground** to clear a GIZMO alarm.

Document status	Commissioning draft; technical and electrical-safety approval required before field use.
Primary audience	Activity coordinators, operators, SC/DPS developers, and authorized electrical personnel.
Current hardware basis	Kria-based GIZMO rack described by DUNE-doc-35392-v2; its implementation defines the authoritative <code>urn:fnal:gizmo</code> OPC UA contract.
Spare basis	Preserved ZedBoard/Zynq GIZMO and DUNE-doc-1805-v12, 25365-v1, and 27482-v2.
Machine contract	The Kria implementation generates the authoritative, hardware-neutral <code>urn:fnal:gizmo</code> OPC UA model; the companion GIZMO-SC/DPS ICD explains it for human review. Any other GIZMO implementation shall conform to the generated contract. Current released model baseline: 1.4.0.
Controlled procedure	Procore: FDC > Administrative > Work Planning and Controls > SOP004 GIZMO. SOP004 authorizes the installation and use of GIZMO. Follow its current approved revision and the applicable work plan.

Contents

1	The one-minute explanation	3
1.1	Rates that must not be confused	3
2	Conceptual connection	4
3	Roles and stop-work boundaries	5
4	Before anyone connects anything	5
4.1	Documents and authorization	5

4.2	Identify the two references	6
4.3	Rack-door reference	6
4.4	Visual and continuity checks	6
5	Connection sequence	6
5.1	Make connections with the rack de-energized	7
5.2	Independent connection verification	7
6	Step-by-step power-up	8
6.1	Pre-power state	8
6.2	Energize and verify	8
6.3	What remains site-controlled	9
7	Commissioning and baseline	9
7.1	Characterization prerequisites	9
7.2	Frequency selection	9
7.3	Threshold selection	10
8	Routine live monitoring	10
8.1	At the beginning of a shift	10
8.2	Minimum live display	10
8.3	Logging format and retention	11
9	Welding and other planned ground shorts	12
9.1	Before the welding window	12
9.2	When the intentional bond is made	12
9.3	Close the welding window	13
9.4	Monitoring coverage and activity state	13
10	Alarm and fault response	13
10.1	Fault-isolation principle	14
11	Normal shutdown and maintenance	15
12	ZedBoard spare unit	15
13	Kria commissioning record and open items	16
14	Shift record template	18
15	Source documents	18

1 The one-minute explanation

The Far Detector needs two electrical reference systems that serve different purposes:

Detector Ground The cryostat steel containment vessel and the metal structures intentionally bonded to it. This is the quiet reference used by detector electronics.

Cavern/Building Ground The bonded infrastructure ground, including the cavern and UFER grounding systems and the grounding conductors used by conventional electrical equipment.

The detector is not simply left floating. An approved safety-ground path connects the two references through the approved *saturable-inductor assembly*. At small AC currents an inductor presents impedance and helps limit noise current. Under a large fault current it saturates and provides a low-impedance path so that protective devices can operate. The monitor is not the safety ground; it only observes the integrity of the grounding arrangement.

GIZMO (Ground Current Impedance MOnitor) injects a small, known AC stimulus between the two ground references. A Pearson current transformer (CT) measures the portion that returns through the intended safety-ground path. If another conductive path appears—a cable shield, pipe, temporary bond, tool, welding return, or accidental contact—some current bypasses the CT path. GIZMO detects the change and reports a lower equivalent impedance or resistance.

Operator note: A low reading does not identify where the short is. It says that another connection exists between the two reference systems. During an intentional welding bond, GIZMO cannot distinguish that approved temporary connection from an accidental one. Work-state information must therefore be recorded alongside the electrical measurement. Work state is an external SC/operations record, not a value supplied by the current GIZMO OPC UA contract.

1.1 Rates that must not be confused

Quantity	Nominal or implemented value	Meaning
Excitation frequency	1.4 kHz	Frequency of the injected <i>sine wave</i> . The legacy source does not describe this as a pulse train.
ADC/DAC rate	Approximately 1 MS/s	Internal waveform generation and digitization rate in the ZedBoard design.
Kria fast-history cadence	1 Hz	Implemented cadence for measurement, alarm, lock-in, thermal, uptime, and sequence snapshots.
Kria platform cadence	0.1 Hz	Implemented cadence for chartable operating-system, storage, process, filesystem, and network counters.

The legacy program automatically loads its 1.4 kHz waveform when the RUN command starts. A 2021 design study suggested approximately 3 kHz for an assumed Far Detector parasitic capacitance, but that was a model, not a commissioned setting. The maintained Kria source defaults to 1436 Hz; the installed value must still be selected from an in-situ frequency sweep and calibration, then frozen in the approved configuration.

The Kria implementation generates one authoritative, hardware-neutral OPC UA contract. Its namespace URI is `urn:fnal:gizmo`; the GIZMO-SC/DPS ICD explains that machine contract for human review. Any other GIZMO hardware implementation shall conform to the generated contract. Site endpoints, certificates, trust lists, roles, and operator authorization remain controlled configuration. The preserved ZedBoard now provides a separate conforming OPC UA endpoint, but it is not a commissioned monitoring

substitute: its resistance remains uncertain until installed-loop calibration is accepted, and `Alarm.Active` is unavailable until the controller/display alarm bit is recovered. Section 12 gives the current readiness limits.

2 Conceptual connection

Figure 1 is intentionally conceptual. It explains the measurement but does not replace the released Far Detector cable schedule or grounding single-line drawing.

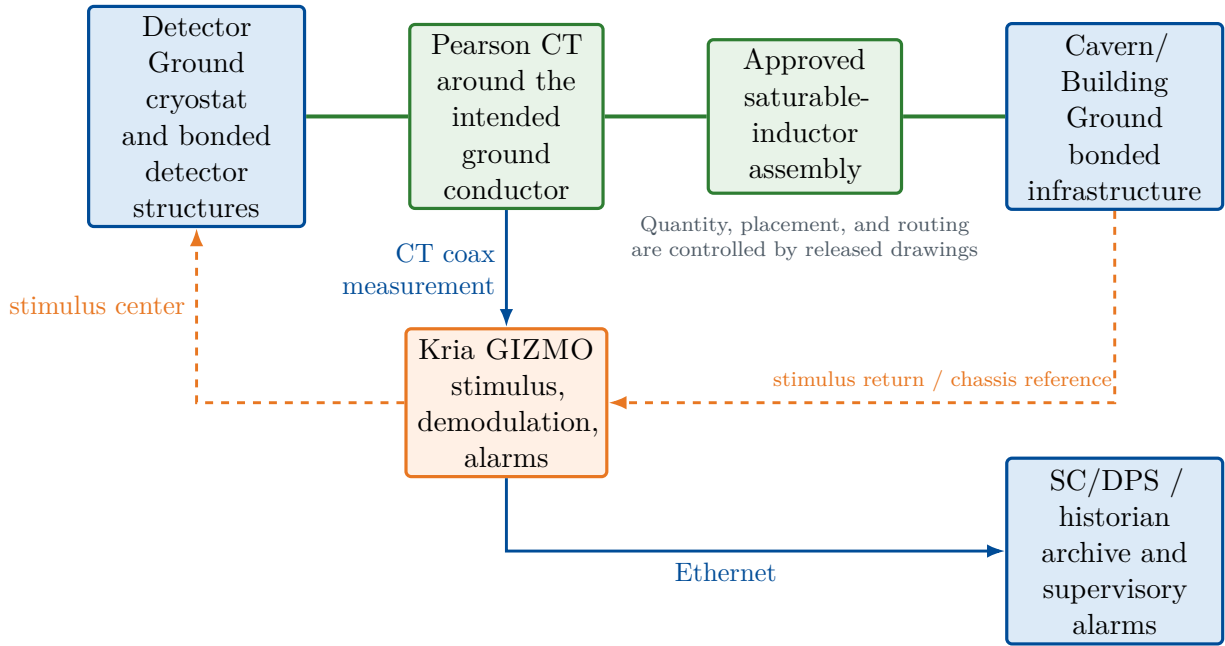


Figure 1: Conceptual measurement topology. The CT must observe the intended return path; any parallel detector-to-infrastructure connection changes the measured signal.

The 2018 Far Site grounding requirements state that the Detector Ground is to remain isolated from other grounding systems except through saturable inductors, require ground-monitoring instrumentation, and call for at least two saturable inductors near the detector isolation transformer with separation greater than six feet. The 2025 Kria ORC package documents a *laboratory* rack containing one MS6984 saturable inductor and one Pearson 2100 CT. The released installation drawing must resolve the final Far Detector quantity and topology; the laboratory ORC drawing must not be copied blindly underground.

3 Roles and stop-work boundaries

Role	Permitted work in this guide
Authorized electrical worker	Installs and verifies fixed grounding conductors, CT, saturable inductors, protection devices, rack bonds, and approved test fixture under the controlled electrical procedure.
GIZMO hardware expert	Verifies chassis, Kria image, analog chain, stimulus amplitude/frequency, calibration, and alarm outputs.
SC/DPS expert	Configures network, timestamps, one-second fast history, bounded edge retention, off-board replication, archive verification, dashboard, external state records, and stale-data alarm.
Activity coordinator	Declares normal/planned-bond/test states, controls the intentional-bond window, records the affected activity, and confirms closeout.
Operator	Performs the released checklist, watches live status, records events, and stops/escalates when acceptance conditions are not met.

STOP / SAFETY: Stop before energizing or injecting a stimulus if a conductor is unidentified, the CT path is bypassed, the saturable-inductor protection differs from the released drawing, the rack safety bond is missing, the Kria image/configuration is unknown, or no responsible electrical/grounding contact is available.

In this guide, **ground-affecting activity** means any activity that can add, remove, or change a conductive path between Detector Ground and Cavern/Building Ground. Examples include welding returns, temporary bonds, cable shields, piping, scaffolding, test jumpers, and changes to grounding or monitoring conductors. Pause and restart decisions depend on the current activity and available monitoring coverage; they are never based on an allowed number of stops per shift.

4 Before anyone connects anything

Complete this section at initial installation and whenever the rack, grounding topology, CT, saturable inductor, firmware, or cryostat connection has changed.

4.1 Documents and authorization

- ☐ Confirm the current approved revision of SOP004 GIZMO, which authorizes installation and use, and the activity-specific hazard analysis.
- ☐ Confirm the released Far Detector grounding single-line, cable schedule, CT orientation, saturable-inductor/MOV assembly drawing, and welding plan.
- ☐ Confirm the Kria rack ORC is released for the installed location and configuration. DUNE-doc-35392 documents the G271 development rack, not automatically the underground installation.
- ☐ Record the equipment IDs: rack, Kria chassis, PDU, CT, saturable-inductor assembly components, AC/DC converter, calibration, software, FPGA image, and configuration hash.
- ☐ Verify that the rack label includes the rack equipment ID and applicable ORC number.
- ☐ Name the electrical work lead, GIZMO expert, SC/DPS contact, activity coordinator, and escalation contact for the shift.

4.2 Identify the two references

An authorized electrical worker shall identify and label:

- the approved *Detector Ground* connection/test point;
- the approved *Cavern/Building Ground* reference/test point;
- the intended safety-ground conductor passing through the CT;
- the detector side and infrastructure side of the saturable-inductor assembly;
- the approved connection point for the stimulus center conductor;
- the rack/chassis bonding point used by the stimulus return.

Important: Do not infer “detector side” from wire color, physical direction, or an old photograph. If the released installation adopts the site convention, Cavern Ground is green and Detector Ground is green with a yellow stripe; both colors remain visual aids only. Verify the released drawing and durable text labels. The ZedBoard reference documents use transformer-secondary and busbar names that may not match the installed Kria labels.

4.3 Rack-door reference

Post a concise, revision-controlled reference on the rack door. It identifies the rack/device and ORC number, both ground references, the normal local alarm indications, **Alarm.Active**, data age and quality, the meaning of **HIGH Z**, current contacts, and the Procore Daily Log location. It also states **Never lift the safety ground**. A link or QR code may point to the current controlled procedure. The rack-door reference is a recognition and navigation aid, not a substitute for that procedure or a definition of SC/DPS response logic.

4.4 Visual and continuity checks

With the applicable circuits in the safe state and LOTO applied where required:

1. ☐ Inspect the rack, CT enclosure, inductor/protection enclosure, connectors, strain relief, and conductor clearances.
2. ☐ Verify the rack and GIZMO chassis are bonded to *Cavern/Building Ground*.
3. ☐ Verify the intended detector safety-ground path is continuous and matches the approved design.
4. ☐ Verify the CT surrounds only the intended conductor(s), is mechanically secure, and is not bypassed by an unmonitored parallel conductor.
5. ☐ Verify no temporary welding lead, shield, pipe bond, test jumper, construction cable, or tool bridges Detector Ground and Cavern/Building Ground.
6. ☐ Verify the alarm beacon is visible and the audible device is appropriate for the work area.

5 Connection sequence

This sequence describes the signal and rack interfaces. Fixed grounding work remains the responsibility of the authorized electrical worker and the controlled installation procedure.

5.1 Make connections with the rack de-energized

1. **Safety-ground assembly.** The authorized electrical worker installs or verifies the approved Detector Ground–CT–saturable-inductor/protection–Cavern/Building Ground path. Do not open this path as part of routine GIZMO work.
2. **CT signal.** Connect the approved coaxial cable from the Pearson CT output to the GIZMO connector designated CT INPUT. Verify both endpoint labels and cable ID.
3. **Stimulus.** Connect the stimulus output using the approved custom cable:
 - the center conductor goes to the designated Detector Ground-side injection point;
 - the outer conductor/return is bonded to the designated rack or Cavern/Building Ground point.

Do not improvise this cable or move the injection point without repeating characterization.

4. **Alarm outputs.** Connect the warning beacon and audible alarm to the labeled GIZMO outputs. Do not use an alarm-power connector as a general auxiliary supply.
5. **Network.** Connect the Kria GIZMO Ethernet port to the assigned detector controls network switch port. Connect the Raritan PDU management port separately if required by the rack drawing.
6. **DC power.** Connect the approved NRTL-listed AC/DC converter to the Kria chassis. DUNE-doc-35392 identifies a 12 VDC, 5 A converter for the development rack. Verify the installed chassis label before connection.
7. **PDU input.** Connect the Raritan PDU only to the approved branch circuit and adapter specified by the installed rack ORC. Check protective-earth continuity before energizing.

STOP / SAFETY: Never perform make-before-break or bypass work on the CT/safety-ground conductor from this guide. The old setup package references a separate CT bypass procedure. That operation requires its own released procedure, tools, hazard controls, and authorized electrical worker.

5.2 Independent connection verification

Before power-up, a second qualified person should perform a point-to-point check against the released drawing. Read each cable label aloud and record initials for:

Connection	Installed by	Verified by
CT output → GIZMO CT input		
Stimulus center → approved Detector Ground-side point		
Stimulus return/chassis → approved Cavern/Building Ground		
Beacon and audible alarm outputs		
Kria and PDU network ports		
12 VDC supply and PDU branch circuit		
Safety path and CT routing match released drawing		

6 Step-by-step power-up

6.1 Pre-power state

The work state used in this section is an external SC/operations record. The current GIZMO OPC UA contract does not publish a work-state selector.

1. ☐ Confirm work state is **COMMISSIONING** or **NORMAL**; no undeclared welding activity is in progress.
2. ☐ Confirm all connection-verification rows above are signed.
3. ☐ Confirm the approved logging and archive plan is available, the Kria has sufficient storage headroom, and every required off-board replica is provisioned for post-boot verification.
4. ☐ Confirm the approved Kria software/FPGA/configuration identifiers are available. If any identifier is unknown, record it as unknown and do not claim the system is commissioned.
5. ☐ Confirm the selected excitation is disabled until the electrical lead authorizes injection.

6.2 Energize and verify

1. Energize the rack PDU from the approved branch circuit. Verify normal PDU status and protective-earth indication.
2. Energize the Kria GIZMO 12VDC supply. Check for normal front-panel power indicators, abnormal sound, odor, heat, or smoke. De-energize immediately if any abnormal condition appears.
3. Wait for the approved Kria boot interval. Confirm `gizmo.target` and the required component services are healthy through the approved operational workflow; record the installed runtime and model versions.
4. Verify OPC UA reachability from the assigned SC/DPS host, resolve `urn:fnal:gizmo` at connection time, and validate device identity. Use the registered hostname, address, certificate, and trust policy from site configuration; do not use historic ZedBoard connection data from old documents.
5. Verify the system clock and data timestamp are UTC and advancing. The legacy display required the measurement time to remain within 10 seconds of the system time; the Kria acceptance criterion is **TBD**.
6. Start the approved acquisition target and confirm the configured excitation waveform, frequency, amplitude limit, and calibration ID before enabling stimulus. Public service names and commands are documented in the maintained Kria runtime; their use still requires the site-approved procedure.
7. With authorization from the electrical lead, enable the configured stimulus only after its waveform, frequency, and amplitude have been approved for the installed topology. Record the actual settings; do not substitute a legacy nominal value.
8. Confirm the local historian records one fast snapshot per second, sequence numbers advance, timestamps are monotonic, platform snapshots arrive every ten seconds, and no parse/quality flags are set.
9. Confirm a plausible stable reading against the most recent commissioned baseline. Do not use a ProtoDUNE, LArTF, MINOS, or laboratory resistance value as the Far Detector baseline.

10. Verify the dashboard shows measurement quality, `ResistanceRange`, resistance or `HIGH Z`, threshold, data age, hardware state, `Alarm.Active`, and `Alarm.Reason`. If an external work state is displayed, verify its source and timestamp. Confirm the SQLite edge store is advancing and every configured replica cursor is healthy.

Expected result: The system is ready for routine monitoring only when the measurement and `Alarm.Active` have Good status, the reported range state matches the commissioned normal state, the threshold and calibration ID are correct, data age is within its approved limit, local storage is active, and the external work state is explicitly recorded.

6.3 What remains site-controlled

The generated GIZMO OPC UA schema defines the canonical namespace, NodeIds, types, meanings, quality and timestamp rules, configuration writes, and operations. The GIZMO-SC/DPS ICD explains that contract for human review. The public Kria runtime implements the present baseline and also defines its own component services, historian, and dashboard. It intentionally does not publish production endpoints, credentials, certificate material, target images, calibration state, network configuration, or authorization policy. The controlled revision of this guide must identify the approved runtime/configuration, endpoint, trust and role policy, exact field commands, rollback, and recovery steps. Legacy commands such as `./GIZMO.elf`, `RUN`, `load_f`, and `set_th` are not Kria commands.

7 Commissioning and baseline

Commissioning establishes what “normal” means for the actual cryostat, grounding conductors, saturable inductors, CT, stimulus cable, and installed firmware. A threshold without a site-specific characterization is not defensible.

7.1 Characterization prerequisites

The legacy characterization procedure requires equipment powered from the detector isolation transformer to be powered down, the transformer primary locked out/tagged out, and detector-ground connections controlled before a decade box is inserted. Perform characterization only under a current approved procedure that explicitly authorizes the installed configuration and test fixture. Do not reproduce the legacy test casually during construction or operation.

Use only an approved resistive/capacitive test fixture. The test plan should obtain:

- open/reference response and noise distribution;
- response to known resistances spanning normal isolation through alarm conditions;
- response to known capacitances representative of the installed detector;
- frequency response over the approved scan range;
- response with representative construction equipment on and off;
- end-to-end verification of beacon, audible alarm, SC/DPS alarm, and logging flags.

7.2 Frequency selection

Select the operating frequency from measured data, not tradition alone. The chosen frequency should:

1. avoid resonances of the installed saturable-inductor and detector-capacitance network;

2. avoid strong coherent facility or switching-noise lines;
3. provide a large and monotonic separation between the open/normal state and known resistive shorts;
4. provide stable I/Q demodulation and acceptable variance over representative work conditions;
5. remain within the approved stimulus-current and touch/noise limits.

Record the sweep, fit, chosen frequency, waveform, amplitude, uncertainty, calibration file/hash, and approver. Changing frequency, amplitude, CT, stimulus routing, saturable inductor, or major cryostat bonding requires review and normally re-characterization.

7.3 Threshold selection

The threshold is a site-specific engineering decision. The approved value must document both its nuisance-alarm margin from the normal distribution and its detection margin for the maximum short resistance that must be found. Do not copy the legacy $100\ \Omega$ example or historic $5\text{--}6\ \Omega$ direct-short reading into the Far Detector configuration without validation.

8 Routine live monitoring

8.1 At the beginning of a shift

1. Confirm the external work-state record uses the approved vocabulary and matches the actual state—for example, normal, welding/planned bond, test, maintenance, or unknown. Verify its source and timestamp; do not treat it as GIZMO data.
2. Confirm data age is below the approved stale limit and the last record time is current UTC.
3. Confirm `Alarm.Active` is False with Good status and review `Alarm.Reason`. If `ResistanceRange` is `InRange`, compare `ResistanceOhm` with the approved baseline band. If Good-quality `OutOfRange/HIGH Z` is the commissioned normal state, accept the range state without inventing or comparing a numeric resistance. Treat `Invalid` or non-good quality as loss of a valid measurement.
4. Confirm frequency, amplitude, calibration ID, firmware/configuration ID, and threshold match the shift reference.
5. Review the previous 12 hours for steps, drift, gaps, or unclosed welding windows.
6. Confirm local historian, free-space guard, replica cursor, and archive status. Record the check in **Procore Daily Log** > **GIZMO** and the shift record.

8.2 Minimum live display

The operator should see, on one screen:

- `ResistanceRange`, resistance or `HIGH Z`, and threshold;
- a time trend with at least the previous 12 hours;
- I, Q, and magnitude or equivalent Kria diagnostics;
- source timestamp, data age, and OPC UA `StatusCode/quality`;

- `Alarm.Active`, `Alarm.Reason`, latch state, and acknowledgement history;
- excitation frequency and the approved controlled configuration;
- hardware/configuration/calibration IDs;
- externally sourced activity state and planned-bond permit/window, with source and timestamp;
- local historian, replica, archive, and network health;

The current GIZMO OPC UA contract does not supply activity state, planned-bond permit/window, alarm-inhibit state, stimulus enabled state, stimulus amplitude, or waveform. If SC displays any of these, it shall use a defined external source and label that source clearly; the display shall not present them as measurements from GIZMO.

8.3 Logging format and retention

The implemented Kria historian uses SQLite schema version 2. Do not use a Procore spreadsheet as the raw data store and do not copy the live SQLite file while its WAL writer is active. Create an immutable archive with the SQLite online-backup API or a checkpointed export, then record its checksum, schema version, runtime version, model digest, and capture-path digest in the controlled archive manifest.

Record class	Implemented cadence / retention	Purpose
Fast snapshot	1 second / 14 days on Kria	Measurement, authoritative alarm, lock-in values, phase, thermal values, uptime, and sequence.
Platform snapshot	10 seconds / 30 days on Kria	Chartable operating-system, storage, process, filesystem, and network counters.
Minute rollup	1 minute / 365 days on Kria	First, last, minimum, maximum, mean, count, and worst OPC UA status.
Transition event	on change / 1825 days on Kria	Alarm, quality, clock, boot, network, service, and historian state changes.

Every stored scalar retains its OPC UA value, `StatusCode`, and source timestamp; the row also records historian receive time and, for fast data, sequence and stream epoch. Raw SDR/ADC frames are excluded. The measured planning rate is approximately 1.48 GB per continuously operating month. The bounded Kria policy occupies about 1.26 GB in replay; a permanent off-board replica may disable age-based pruning while retaining the free-space guard and independent backup requirements.

Replication is cursor-based, bounded, transactional, and idempotent. It transfers inserts and updated rollups, never deletion. The board-local store is the authoritative edge buffer; each off-board replica and the site Ignition historian are independent downstream records with their own retention and restore acceptance.

The long-term Ignition historian uses the PostgreSQL direction established for Slow Controls. Production hosting, network access, retention, backup, recovery, security, and service ownership follow the SC-Computing ICD and site operations. This does not replace the bounded board-local SQLite edge buffer or its independent archive and recovery role.

The **Procore Daily Log > GIZMO** section should contain human summaries and controlled events:

- automatically generated 6-, 12-, or 24-hour plots and statistics;
- shift start/end checks;

- welding-window start/end and permit/work-package reference;
- alarm events, acknowledgements, investigation, and disposition;
- links to the immutable archive, checksum, configuration, and calibration.

9 Welding and other planned ground shorts

Welding or another approved temporary bond may intentionally connect Detector Ground to Cavern/Building Ground. During that interval, **Alarm.Active** or a low reading may be expected, but the intentional connection also masks any second accidental short. The activity window must therefore be controlled and time-correlated with the GIZMO data and its external activity-state record.

The uppercase event labels below are descriptive human-log names, not GIZMO OPC UA nodes. The controlled procedure and SC/DPS configuration approve the final activity-state and event vocabulary.

9.1 Before the welding window

1. The activity coordinator confirms the approved welding or bonding procedure, work package, location, work lead, expected grounding/bonding arrangement, and start/end authorization.
2. The electrical/grounding lead confirms that the welding return and temporary bonds match the approved plan. GIZMO does not authorize or validate welding-current routing.
3. For the approved pre-work stabilization interval, verify Good measurement quality, **Alarm.Active** is False with Good status, and the commissioned normal range/baseline state. The interval and acceptance criterion must come from the controlled procedure; this guide does not assign a duration.
4. Record a **WELDING_PREP** event with UTC time, work-package ID, location, work lead, expected bond configuration, and expected duration.
5. Confirm continuous one-second fast-history capture and a healthy edge buffer. Do not stop acquisition because a short is expected.
6. If audible/visual annunciation must be inhibited to conduct the work, use only the approved SC/DPS alarm-management method. Preserve raw **Alarm.Active**, its **StatusCode** and timestamp, and an unmistakable externally sourced **ALARM_INHIBITED** record. Do not disconnect the CT, stimulus, safety-ground path, or monitor to silence an alarm.

9.2 When the intentional bond is made

1. Record **WELDING_ACTIVE** at the actual time the bond/return becomes effective.
2. Confirm GIZMO sees the expected transition. If no change is observed when the approved topology predicts one, stop and ask the grounding/GIZMO expert to verify the measurement path.
3. Keep the dashboard visible and monitor data age, hardware health, and logging continuity. Impedance alone is not a valid isolation indicator while the intentional short exists.
4. Record any additional temporary bonds, cable changes, moves, pauses, or work-area changes as timestamped events.
5. Treat a loss of valid GIZMO data as loss of monitoring coverage and follow Section 9.4.

9.3 Close the welding window

1. The work lead confirms the activity has stopped and all temporary leads/bonds identified in the work plan have been removed or placed in their specified final state.
2. Record `WELDING_BOND_REMOVED` at the actual removal time.
3. Verify that `Alarm.Active`, measurement quality, `ResistanceRange`, and any numeric resistance return to the commissioned normal state within the approved recovery window.
4. Require the approved stable post-work interval with no non-good quality and no departure from the approved normal range/baseline criterion. The duration and criterion must come from the controlled procedure; this guide does not assign them.
5. If the reading does not recover, keep the work area controlled. Do not begin by removing the safety ground. Inspect the newly installed work, temporary leads, shields, piping, scaffolding, tools, and recent cable connections using the approved fault-isolation plan.
6. Record `WELDING_CLOSED` only after the electrical/grounding lead and activity coordinator accept the post-work state. Attach the before/after plot to the shift record and **Procore Daily Log > GIZMO** entry.

STOP / SAFETY: An intentional welding short is not a nuisance alarm to be ignored. It is a declared alternate system state. If the activity state is unknown, the planned window has expired, or the physical bond configuration does not match the approved plan, treat `Alarm.Active` or the unexpected measurement transition as a possible ground fault.

9.4 Monitoring coverage and activity state

Loss of valid monitoring does not create the same operational consequence in every state. Use the approved work plan and the following boundary:

Current state	Required response
No ground-affecting activity is active	Raise and record the coverage alarm; preserve data; begin recovery; escalate if the condition persists or recurs. Unrelated activity does not pause solely because the GIZMO report is stale.
Ground-affecting activity is active and an approved alternate monitor is available	Continue only as permitted by the approved alternate-monitor procedure, and record the alternate source and interval.
Ground-affecting activity is active and no approved alternate monitor is available	Pause the ground-affecting activity under the approved work plan until valid coverage or an approved alternative is restored.

Do not establish a quota for monitoring-related pauses or an allowed number of stops per shift. If coverage loss persists or recurs, notify both the GIZMO expert and the SC/DPS expert and record the disposition.

10 Alarm and fault response

Observation	Meaning	Immediate operator response
<code>Alarm.Active=True</code> with Good status during NORMAL	The device's authoritative composite alarm logic reports a possible new conductive path or other configured alarm condition	Stop new ground-affecting activity; record UTC time and <code>Alarm.Reason</code> ; notify the activity coordinator and grounding/GIZMO expert; preserve data; inspect recent changes under the approved fault plan.
<code>Alarm.Active=True</code> with Good status during a declared planned-bond window	The intentional bond may be the expected cause, but additional shorts are masked	Verify the active work package, bond configuration, location, and time window. Continue quality, freshness, and history monitoring. If the declared state is wrong or expired, handle the report as an unexpected fault.
Good <code>OutOfRange/HIGH Z</code> range with <code>Alarm.Active False</code>	The resistance is above the validated numeric presentation range; this is a valid range state, not a numeric value or data-quality fault	Display and record HIGH Z. Do not substitute 500, 999, zero, or another numeric value. Confirm that HIGH Z is consistent with the commissioned normal state.
Sudden measurement step while <code>Alarm.Active=False</code>	Topology or capacitive coupling may have changed without crossing the device's composite alarm condition	Record the event, identify simultaneous activity, compare I/Q and magnitude, and ask the grounding expert whether a new baseline or characterization is required. Do not create a second alarm decision in SC/DPS by recomputing the threshold logic.
Stale timestamp, missing samples, sequence gaps, or non-good <code>Alarm.Active</code> status	Monitoring coverage is lost or degraded; a retained value is not a fresh clear state	Raise a distinct coverage alarm, preserve the last value with its <code>StatusCode</code> , follow Section 9.4, and recover the logger without erasing the local file. Escalate persistent or recurring faults to both the GIZMO and SC/DPS experts.
Invalid resistance range, implausible value, clipping, I/Q discontinuity, or frequency mismatch	Analog chain, CT, stimulus, configuration, or calibration problem	Mark the measurement invalid; do not reinterpret it as good isolation. Check configuration ID and hardware status; escalate to the GIZMO expert.
<code>Alarm.Active</code> does not activate during an authorized test	End-to-end protection/notification failure	Do not release to routine monitoring. Verify threshold, test fixture, beacon, audible output, SC/DPS path, and CT/stimulus topology.

STOP / SAFETY: Never lift the safety ground in response to an alarm, stale report, invalid measurement, or failed recovery.

10.1 Fault-isolation principle

Look first at what changed immediately before the transition. Use activity records to divide the search by area or work package. Disconnect or isolate candidate paths only under the controlling procedure and

with the responsible system owner. Never use GIZMO as permission to disconnect protective grounding or defeat electrical safety.

11 Normal shutdown and maintenance

1. Record the reason, UTC time, external activity state, `Alarm.Active`, measurement quality/range, threshold, configuration ID, and last healthy sample.
2. If ground-affecting activity is active, follow Section 9.4 before taking the monitor out of service.
3. Verify the last historian write and every required replica cursor. If an archive is required, create a consistent online backup or checkpointed export, calculate its checksum, and record the storage reference.
4. Disable the excitation using the approved Kria control: **TBD**. Confirm stimulus-off using the independent indication defined by the controlled procedure; the current GIZMO OPC UA contract does not publish a stimulus-enabled node.
5. Stop the acquisition service using the approved Kria method: **TBD**.
6. De-energize the Kria chassis and rack PDU only when authorized by the rack procedure.
7. Disconnect signal, network, or DC cables only after power is off and the work is released.
8. Leave the permanent safety-ground and saturable-inductor path in its approved state. Any work on it requires the separate electrical procedure.

12 ZedBoard spare unit

The preserved ZedBoard is the GIZMO spare. It now runs its own native OPC UA server as a non-owning sidecar to the preserved controller and exposes the same `urn:fnal:gizmo` model as the Kria. It has a distinct endpoint and device identity; it does not proxy or control the Kria.

Item	Current spare status
Machine contract	Model baseline 1.4.0 with the common namespace, string NodeIds, datatypes, units, access metadata, and explicit non-good status for unavailable functions.
Measurement	Installed-loop calibration is not yet accepted. Impedance values therefore report <code>UncertainLastUsableValue</code> ; capacitance and phase are unsupported.
Authoritative alarm	<code>Alarm.Active</code> reports <code>BadDataUnavailable</code> until the actual controller/display alarm-return bit is recovered. A Boolean value accompanied by that status is not an authoritative clear alarm.
Controls	Anonymous access is read-only. The separately gated authenticated <code>Threshold0hm</code> write accepts only the ZedBoard subset 0–1023 ohms. Other writes and all operation methods are unsupported.
Network security	The current maintenance endpoint uses OPC UA SecurityPolicy None, including for username/password control. It is restricted to the isolated maintenance network and must not be routed or bridged to an untrusted network.
Readiness	Not accepted as a substitute for Kria ground-short reporting or SC/DPS delivery until calibration, alarm-bit recovery, identity, non-interference, restart, network-loss, security, display comparison, and end-to-end acceptance pass.

Important: Calling the ZedBoard a spare identifies its intended role; it does not declare it ready. Operators shall not select it as the monitoring or DPS source until its controlled acceptance record is complete.

13 Kria commissioning record and open items

The following fields must be completed before this draft can become a field procedure.

Controlled item	Approved value / reference
Underground rack ID and released ORC	TBD
Far Detector grounding single-line and cable schedule	TBD
Saturable-inductor quantity, IDs, protection, and placement	TBD
Pearson CT ID, orientation, cable ID, and transfer factor	TBD
Detector-side stimulus connection point	TBD
Cavern/Building Ground stimulus return point	TBD
Kria hostname, OPC UA endpoint, certificate/trust policy, and network owner	TBD
Kria OS, FPGA image, application, and configuration IDs	TBD
Approved runtime version; startup, enable, disable, restart, and recovery commands	TBD

Controlled item	Approved value / reference
Boot time, update cadence, stale-data limit, and recovery time	TBD
Excitation waveform, frequency, amplitude, and tolerance	TBD
Calibration file/hash, validity range, uncertainty, and owner	TBD
Normal baseline band and alarm threshold	TBD
Authorized alarm-test fixture and procedure	TBD
SC/DPS endpoint, roles, selected fields, status bits, and alarms	TBD
Historian schema 2 archive, local retention, checksum, and restore record	TBD
Off-board replica/archive destination, retention, and owner	TBD
External activity-state/planned-bond interface and authorized alarm-inhibit method	TBD
Approved pre-work, post-work, recovery, and monitoring-coverage criteria	TBD
Applicable SOP004 revision and activity-specific work-plan reference	TBD
ZedBoard spare calibration, alarm-bit recovery, security, and acceptance record	TBD
Rack-door reference revision and Procore Daily Log > GIZMO location	TBD
Shift electrical lead, GIZMO expert, SC/DPS contact, activity coordinator, escalation contact, and emergency contacts	TBD

14 Shift record template

Date / shift / operator
 Rack / chassis / configuration ID
 Applicable ORC / rack-door reference
 revision
 Calibration ID / threshold
 Excitation waveform / frequency /
 amplitude
Alarm.Active / reason / quality / range
 state at shift start
 Baseline or approved HIGH Z state at shift
 start
 Data age / historian / replica / archive
 status
 External activity state / work package
 Planned-bond windows and work lead
 Electrical lead / GIZMO expert
 SC/DPS contact / activity coordinator /
 escalation contact
 Alarms, steps, gaps, or interventions
 Alarm / quality / range / baseline at shift
 end
 Archive reference and checksum
 Summary plot / Procore Daily Log >
 GIZMO record
 Open issues and handoff

15 Source documents

1. DUNE-doc-35392-v2, *ORC Documents for DUNE Impedance Monitor Rack in FNAL HEERC G271*. Current Kria rack hardware basis; laboratory/development configuration.
<https://docs.dunescience.org/cgi-bin/private/ShowDocument?docid=35392>
2. DUNE-doc-1805-v12, *GIZMO: Ground Current Impedance MONitor*. Legacy engineering note, operations, setup instructions, diagrams, and schematics.
<https://docs.dunescience.org/cgi-bin/private/ShowDocument?docid=1805>
3. DUNE-doc-25365-v1, *LArTF Impedance Monitor System for 2x2*. 2022 engineering note, characterization, and operating tips.
<https://docs.dunescience.org/cgi-bin/private/ShowDocument?docid=25365>
4. DUNE-doc-27482-v2, *MINOS Impedance Monitor System*. Legacy SD-card, chassis, calibration, PMOD, and testing material.
<https://docs.dunescience.org/cgi-bin/private/ShowDocument?docid=27482>
5. DUNE-doc-285-v11, *DUNE Grounding Plan*; file *LBNF/DUNE Far Site Detector Grounding System Requirements*, April 8, 2018.
<https://docs.dunescience.org/cgi-bin/private/ShowDocument?docid=285>
6. D. Subedi, *Ground Impedance Monitor Technical Report*, 2021. ZedBoard architecture and Far Detec-

tor frequency study.

https://dsubedi.info/data/I1/FNAL_2021_GIZMo_paper.pdf

7. V. Cerrone, *DUNE ND-LAr 2x2 Demonstrator Tests*, 2022. Legacy SSH, InfluxDB, and Grafana readout implementation.
[Fermilab Indico report attachment](#)
8. Public legacy client source, `gizmo_control`.
https://github.com/vanessacerrone/gizmo_control
9. Public GIZMO Kria runtime, generated OPC UA contract, reference implementation, historian, dashboard, and ZedBoard-spare status. The generated schema is the machine-interface authority.
<https://github.com/marroyav/GIZMo>
10. Public GIZMO-SC/DPS ICD and dual-platform DCS intake package; the ICD explains the generated contract for human review.
<https://github.com/marroyav/gizmo-icd>
11. Public GIZMO historian archive schema and verification tools.
<https://github.com/marroyav/gizmo-data-archive>
12. Public GIZMO Ignition project and history-migration tooling.
<https://github.com/marroyav/gizmo-ignition>

Important: Several source documents describe prototypes or installations other than the DUNE Far Detector. Their values are clearly labeled as legacy or proposed in this guide. The current released drawings, Kria software manual, site characterization, and approved work controls take precedence.